



**Old tech:** The basic design of carbon-fibre blades has remained largely unchanged since the 1990s, but improved athleticism is narrowing the gap with able-bodied athletes, says former Paralympic champion Heinrich Popow.

Pics: JONATHAN SELVARAJ

the 2010s by Paralympic 400m champion Oscar Pistorius, but its design has remained virtually unchanged since engineer Van Phillips created it in 1984.

And it is not the only vintage design still in use. "The current world records in the 100m, 200m and 400m (until last year) in the T64 category are held by Johannes Floors. He's using blades that were first introduced in 1992. That design is older than he is!" says Heinrich Popow, the 2012 London Paralympics 100m champion (T42 category, later merged into T63), who himself competed with Ottobock blades.

Now an equipment expert with Ottobock, Popow points out a paradox: while blades for elite athletes appear futuristic, everyday prosthetics are far more advanced. "For regular users, prosthetics have

come a long way. You have microprocessors, exoskeletons, crazy stuff. For daily wear, you want all the support you can get – sensors, batteries, and AI algorithms to predict your gait. But sport is different. Sport is about overcoming weakness," he says.

**For athletes, the options remain limited. One reason is mechanical.**

The J-shape is near perfect: it acts like a spring, absorbing and releasing energy with every stride – exactly what sprinting and jumping demand.

Another reason is regulatory.

World Para Athletics (WPA) imposes strict limits. Each T64 athlete is given a Maximum Allowable Standing Height (MASH) during classification. If blades make them taller, they cannot compete. "You also can't just debut a new design," says Popow. "The WPA says any blades being used have to be in use for at least nine months."

It is not that manufacturers lack innovation, but that radical designs are restricted. "If it were allowed, you could absolutely create something far superior. But then you might need to put a motor or

« It's not the technology that drives the human – it's the human that drives technology – Heinrich Popow »